Extrema and the Mean Value Theorem

ANSWER KEY

Section 1: Critical Points & First Derivative Test

1 What are the critical points of $f(x) = x^3 - 3x^2 - 9x + 5$?

a. $x = -1$ **and** $x = 3$ ✓

b. $x = 1$ and $x = -3$

c. $x = 0$ and $x = 3$

d. $x = -1$ only

$f'(x) = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x - 3)(x + 1) = 0 \Rightarrow x = -1, 3.$

2 By the First Derivative Test, what type of extremum does $f(x) = x^3 - 3x^2 - 9x + 5$ have at $x = -1$?

a. **Local maximum because f' changes from positive to negative** ✓

b. Local minimum because f' changes from negative to positive

c. Inflection point

d. No extremum

For $x < -1$, $f'(x) > 0$, and for $-1 < x < 3$, $f'(x) < 0$. Since f' changes from positive to negative, $x = -1$ is a local maximum.

3 Can a critical point occur at a value where $f'(x)$ is undefined?

a. **Yes, provided x is in the domain of f** ✓

b. No, f' must always equal zero

c. Only if $f''(x) = 0$

d. Only for rational functions

A critical point of f is any point c in the domain of f where $f'(c) = 0$ OR $f'(c)$ does not exist (such as cusps and corners).

Section 2: Mean Value Theorem (MVT)

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4 What are the conditions required to apply the Mean Value Theorem on $[a, b]$?

a. **Continuous on $[a, b]$ and differentiable on (a, b)** ✓

b. Differentiable on $[a, b]$ and $f(a) = f(b)$

c. Continuous on (a, b) only

d. $f'(c) = 0$ on $[a, b]$

The MVT requires that f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b).

5 Find the value of c guaranteed by the MVT for $f(x) = x^2$ on the interval $[0, 4]$.

a. 2 ✓

b. 1

c. $\sqrt{2}$

d. 2.5

Average rate of change $= \frac{f(4) - f(0)}{4 - 0} = \frac{16 - 0}{4} = 4$. $f'(c) = 2c = 4 \Rightarrow c = 2$.

6 Why does the MVT NOT apply to $f(x) = |x|$ on $[-2, 2]$?

a. **f is not differentiable at $x = 0$** ✓

b. f is not continuous on $[-2, 2]$

c. $f(-2) \neq f(2)$

d. The interval is not symmetric

$f(x) = |x|$ has a sharp corner at $x = 0 \in (-2, 2)$, violating the differentiability condition.

Section 3: Rolle's Theorem

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7 What additional hypothesis does Rolle's Theorem require compared to the MVT?

a. $f(a) = f(b)$ ✓

b. $f'(a) = f'(b)$

c. $f''(x) = 0$

d. $f(a) = 0$

Rolle's Theorem adds the condition that the endpoints have equal function values: $f(a) = f(b)$, guaranteeing at least one c where $f'(c) = 0$.

8 Verify Rolle's Theorem for $f(x) = x^2 - 5x + 4$ on $[1, 4]$. What is the value of c?

a. 2.5 ✓

b. 2.0

c. 3.0

d. 1.5

$f(1) = 0$ and $f(4) = 0$. $f'(x) = 2x - 5 = 0 \Rightarrow c = 2.5 \in (1, 4)$.

9 If a differentiable function has 3 distinct real zeros, what is the minimum number of zeros its derivative $f'(x)$ must have?

a. 2 ✓

b. 1

c. 3

d. 0

By Rolle's Theorem, between each pair of consecutive zeros there must be at least one zero of $f'(x)$. For 3 zeros, there are 2 intervals, so at least 2 zeros of f'.

Section 4: Absolute Extrema on Closed Intervals (Candidates Test)

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10 Find the absolute maximum value of $f(x) = 2x^3 - 3x^2 - 12x + 1$ on $[-2, 3]$.

a. 8 ✓

b. 1

c. -19

d. -8

Critical points: $f'(x) = 6x^2 - 6x - 12 = 6(x - 2)(x + 1) = 0 \Rightarrow x = -1, 2$. Candidates: $f(-2) = -3, f(-1) = 8, f(2) = -19, f(3) = -8$. Absolute maximum is 8.

11 What is the absolute minimum value in the problem above?

a. -19 ✓

b. -8

c. -3

d. 1

From the candidates test above, the lowest evaluated value is $f(2) = -19$.

12 Where can the absolute extrema of a continuous function on a closed interval $[a, b]$ occur?

a. **At critical points inside** (a, b) **or at the endpoints** a, b ✓

b. Only at endpoints

c. Only where $f'(x) = 0$

d. Anywhere on the interval

By the Closed Interval Method (Candidates Test), absolute extrema must occur either at interior critical points or at the boundary endpoints.

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